PUBLIC STRUCTURED CV

Daniel Ari Friedman, PhD

Computational biologist, Active Inference researcher, cognitive security researcher, educator, and artist | California, USA, Earth

Verify	**resume/verify.html** QR target, public hashes, source files, and artifact sizes.
Source	**manifest sha256 494b4b2472077fc2** resume/source.json + canonical data exports
Output	**json sha256 ce4a7ad792186b00** data/resume.json resume/resume.pdf

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resume/verify.html

Curated Works	Software Rows	Scholar Metrics	Resume Records
154 data/works.json	**88** data/software.json	**764** Scholar profile	**119** verify page
Source Hash **494b4b247207** source manifest	**JSON Hash** **ce4a7ad79218** data/resume.json	**PDF Hash** **verify page** resume/verify.html	**Footer QR** **every page** resume/verify.html

Source-to-Output Flow

Inputs	Generated JSON	Public PDF + Verification
resume/source.json data/works.json data/software.json source sha256 494b4b2472077fc2	**data/resume.json** 191,673 bytes json sha256 ce4a7ad792186b00	**resume/resume.pdf** resume/verify.html PDF hash posted on verification page

Section Counts

Education	**4**
Experience	**16**
Awards	**12**
Conferences	**17**
Media / Outreach	**29**
Service	**27**
Art Uses	**14**

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Email danielarifriedman@gmail.com; daniel@activeinference.institute	**Sites** https://danielarifriedman.com/; https://cognitivesecurity.us
GitHub https://github.com/docxology	**LinkedIn** https://www.linkedin.com/in/danielarifriedman/
Scholar https://scholar.google.com/citations?user=DXjPFtYAAAAJ	**ORCID** 0000-0001-6232-9096
Keybase / ENS docxology / docxology.eth	**Art** https://www.flickr.com/photos/daniel_friedman/

Curated Works	Software Rows	Scholar Citations
154 from data/works.json	**88** 56 owned + 32 AII	**764** h 15 / i10 17 as of 2026-05-16

Variant: full. Generated: 2026-06-07T22:43:03Z. Current public metrics: 154 curated works; 88 catalogued software repositories (56 owned + 32 AII); 764 Google Scholar citations, h-index 15, i10-index 17 as of 2026-05-16.

Summary

Multidisciplinary researcher and open-source builder working across social insect biology, Active Inference, cognitive security, computational frameworks, education, and art.

Education (4)

2019-2023 **Postdoctoral Fellow | University of California, Davis | Davis, CA, USA**

- Co-mentors: Prof. Brian Johnson (UC Davis) and Prof. Tim Linksvayer (Arizona State University).

- NSF Postdoctoral Research Fellowship in Biology (PRFB), award DBI-2010290; NSF budget period 2020-2022 with affiliation extending to 2023.

- Genomics, epigenomics, transcriptomics, and behavior in eusocial insects, mainly ants and bees.

- Science research, mentoring, participation, service, and organizing events and communities online.

Source: NSF DBI-2010290 https://grantome.com/grant/NSF/DBI-2010290 | UC Davis https://www.ucdavis.edu/

2014-2019 **PhD | Stanford University | Stanford, CA, USA**

- Advisor: Professor Deborah M. Gordon, Biology.

- Research, teaching, service, and coursework from Summer 2014 to Summer 2019.

- Thesis topics: ant ecology, collective behavior, brain gene expression, and neurometabolism.

- Stanford Graduate Fellow (Urbanek Family Fellowship), 2014-2017.

- Co-organizer of the Stanford Complexity Group, 2016-2019.

- Stanford Neuroscience NeuroChoice Grant for neurometabolism in ants, 2019, $25,000.

- Stanford Systems Biology Seed Grant for gene expression and behavior in ants, 2018, $25,000.

- Theodore Roosevelt Memorial Award, American Museum of Natural History, 2016, $2,500.

- Lewis and Clark Fund grant, American Philosophical Society, 2016, $4,000.

- NSF GRFP Honorable Mention, 2015 and 2016.

Source: Stanford dissertation http://purl.stanford.edu/pb813wm1484

2010-2014 **BSc in Genetics | University of California, Davis | Davis, CA, USA**

- BSc in Genetics, University of California, Davis.

- Genetics Department Citation, Highest Honors, 2014; 3.74 GPA overall.

- Campus Community Service Silver Award, 2014.

- Stanley and Jacqueline Schilling Award for Service, 2014.

- Completed the Population Biology Graduate Group one-year base PhD curriculum, 2013.

- Co-founder of The Aggie Transcript, an undergraduate biology journal, 2014.

- Victoria Finnerty Award for Research in Drosophila, 2013.

- Elected member of Phi Beta Kappa, UC Davis Chapter, 2014.

- Co-organizer of the Chess Club at UC Davis, 2012-2014.

- Research Scholars Program in Insect Biology, 2011-2014.

- Provost's Undergraduate Fellowship for Drosophila research, 2011, $2,500.

Source: UC Davis https://www.ucdavis.edu/

2006-2010 **High School | Crystal Springs Uplands School | Hillsborough, CA, USA**

- Crystal Springs Uplands School.

Source: Crystal Springs Uplands School https://www.crystal.org/

Experience (16)

2026-ongoing **Del Norte Triplicate | Typographical Editor**

Community Organization

- Typographical editing for The Triplicate newspaper of Del Norte County and Crescent City.

Source: The Triplicate https://www.triplicate.com/

2021-ongoing **Active Inference Institute | Co-founder, Participant, Officer, President, Treasurer**

Community Organization

- Co-founded the Active Inference Institute; current President and Treasurer.

Source: All officers https://www.activeinference.institute/officers | All https://activeinference.org/

2019-2024 **Complexity Adventures | Co-founder, Facilitator, Participant, Organizer**

Community Organization

- Organized activities and community around learning and applying complexity science.

Source: Complexity Adventures profile https://www.complexityadventures.com/organizers/2

2019-ongoing **COGSEC | Researcher**

Community Organization

- Research facilitation and participation on initiatives related to cognitive security.

Source: COGSEC https://www.cogsec.org/

2014-2019 **Stanford Complexity Group | Organizer, Participant**

Community Organization

- Organized the Stanford Complexity Group while it was active.

Source: Stanford https://www.stanford.edu/ | verification https://danielarifriedman.com/resume/verify.html

2025-2026 **Stealth Startup | Operator**

Startup

- Systems modeling and related work to be public later.

Source: verification https://danielarifriedman.com/resume/verify.html

2023-2024 **Stealth Startup | Co-founder, Operator**

Startup

- Stealth Active Inference startup.

Source: verification https://danielarifriedman.com/resume/verify.html

2019-2023 **Remotor | Co-founder, Researcher**

Startup

- Education, online event planning, and consulting for remote teams.

Source: verification https://danielarifriedman.com/resume/verify.html

2017-2019 **fm.analytics | Co-founder, Researcher**

Startup

- Predictive analytics and dashboard visualizations with Prof. Glenn Magerman at KU Leuven.

Source: verification https://danielarifriedman.com/resume/verify.html

2017-2019 **Cryptoptera | Co-founder, Researcher**

Startup

- Three-person cryptocurrency research group.

Source: verification https://danielarifriedman.com/resume/verify.html

2019-2023 **Johnson Lab / Linksvayer Lab | Mentor, Researcher**

Research

- Postdoctoral position during NSF fellowship with Brian Johnson at UC Davis and Tim Linksvayer at Arizona State University.

Source: Johnson Lab https://johnsonlab.ucdavis.edu/ | UC Davis https://www.ucdavis.edu/

2014-2019 **Gordon Lab | Mentor, Researcher**

Research

- Graduate research, field work, wet lab, bioinformatics, and mentoring of high schoolers, undergraduates, graduate students, and non-academics.

Source: Gordon Lab https://web.stanford.edu/~dmgordon/

2010-2014 **Kopp Lab | Researcher**

Research

- Undergraduate research assistant with Prof. Artyom Kopp in Ecology and Evolution at UC Davis, studying non-coding DNA elements in the evolution of the Drosophila foreleg sex comb.

Source: UC Davis Biology https://biology.ucdavis.edu/

2014-2019 **Pedicab People Movers | Operator**

Pedicab

- Operated a pedicab/rickshaw at events including music festivals in California and downtown areas in Davis and San Jose.

Source: verification https://danielarifriedman.com/resume/verify.html

2015-2016 **Stanford University | Teacher, Organizer, Graduate Teaching Assistant**

Teaching

- Graduate Teaching Assistant for undergraduate and graduate courses: General Biology: Cell and Molecular Biology (2015) and Ethics of Evolutionary Biology (2016).

Source: Stanford Biology https://biology.stanford.edu/

2014 **UC Davis | Teacher, Organizer, Course Co-developer**

Teaching

- Student teacher and developmental biology course co-developer with Prof. Susan Keen at UC Davis.

Source: UC Davis Biology https://biology.ucdavis.edu/

Awards and Fellowships (12)

Year	Award	Details
2020	**NSF Postdoctoral Research Fellowship in Biology** Source: NSF DBI-2010290 https://grantome.com/grant/NSF/DBI-2010290	Award DBI-2010290; NSF budget period 2020-2022.

Year	Award	Details
2014	**Stanford Graduate Fellow, Urbanek Family Fellowship** Source: Stanford Graduate Fellowship https://vpge.stanford.edu/fellowships-funding/sgf	Stanford University, 2014-2017.
2019	**Stanford Neuroscience NeuroChoice Grant** Source: Stanford Neurosciences Institute https://neuroscience.stanford.edu/	Neurometabolism in ants, $25,000.
2018	**Stanford Systems Biology Seed Grant** Source: Stanford Systems Biology https://sysbio.stanford.edu/	Gene expression and behavior in ants, $25,000.
2016	**Theodore Roosevelt Memorial Award** Source: AMNH Theodore Roosevelt Grants https://www.amnh.org/research/richard-gilder-graduate-school/academics-and-research/fellowships-and-grants/theodore-roosevelt-memorial-grants	American Museum of Natural History, $2,500.
2016	**Lewis and Clark Fund grant** Source: APS Lewis and Clark Fund https://www.amphilsoc.org/grants/lewis-and-clark-fund-exploration-and-field-research	American Philosophical Society, $4,000.
2015	**NSF GRFP Honorable Mention** Source: NSF GRFP https://www.nsfgrfp.org/	Honorable Mention in 2015 and 2016.
2014	**UC Davis Genetics Department Citation** Source: UC Davis Genetics https://genetics.ucdavis.edu/	Highest Honors.
2014	**UC Davis service awards** Source: UC Davis https://www.ucdavis.edu/	Campus Community Service Silver Award and Stanley and Jacqueline Schilling Award for Service.
2013	**Victoria Finnerty Award for Research in Drosophila** Source: UC Davis Genetics https://genetics.ucdavis.edu/	UC Davis.
2014	**Phi Beta Kappa** Source: Phi Beta Kappa https://www.pbk.org/	Elected member, UC Davis Chapter.
2011	**Provost's Undergraduate Fellowship** Source: UC Davis Undergraduate Research Center https://urc.ucdavis.edu/puf	Drosophila research, $2,500.

Works and Publications (154)

#	Year	Type	Title and Venue	Link
001	2026	Paper	**A template/ approach to Reproducible Generative Research: Architecture and Ergonomics from Configuration through Publication | Zenodo**	10.5281/zenodo.19139090
002	2026	Series	**Active Inference Journal — 500+ videos on Active Inference with transcripts | YouTube**	https://video.activeinference.institute/
003	2026	Paper	**Before Pragmatism Had a Name: Blake's "America A Prophecy" Anticipates American Anticipatory Epistemology | Zenodo**	10.5281/zenodo.18807970
004	2026	Paper	**The Golden Compass and the Lunar Flux: William Blake, Bimetallic Meta-stability, and the Architecture of Value | Zenodo**	10.5281/zenodo.19335196
005	2026	Paper	**Cognitive Integrity Framework: Formal Foundations for Multiagent Security (Part 1: Theory) | Zenodo**	10.5281/zenodo.18364119
006	2026	Series	**Personal YouTube — 200+ livestreams: drawings, Synergetics, paper discussions | YouTube**	https://www.youtube.com/@danielarifriedman/playlists
007	2026	Paper	**The Doors of Perception are the Threshold of Prediction: Active Inference and William Blake's Theory of Seeing | Zenodo**	10.5281/zenodo.18600041
008	2025	Presentation	**5th Applied Active Inference Symposium — Abstract Book | Presentation**	10.5281/zenodo.17555266
009	2025	Paper	**Adaptive Basic Income (AuBI): Integrating AI, Decentralized Infrastructure, and Active Inference | Zenodo**	10.5281/zenodo.17228945
010	2025	Paper	**CEREBRUM: Case-Enabled Reasoning Engine with Bayesian Representations for Unified Modeling | Zenodo**	10.5281/zenodo.15170907
011	2025	Paper	**Computational Complexity and Energetics of the Ant Stack | Zenodo**	10.5281/zenodo.17238736
012	2025	Paper	**EvoJump: Stochastic Modeling of Evolutionary Ontogenetic Trajectories | Zenodo**	10.5281/zenodo.17229924
013	2025	Paper	**Increasing the Accessibility and Applicability of Active Inference | Zenodo**	10.5281/zenodo.15061667
014	2025	Paper	**MDKV: A Multitrack Markdown Container for Structured, Portable Documents | Zenodo**	10.5281/zenodo.16790554
015	2025	Paper	**Markdown Decision Process: A Framework for Probabilistic Document Analysis | Zenodo**	10.5281/zenodo.17244386
016	2025	Paper	**On Cognitive Art & Science: Toward Wholeness From Both Sides | Zenodo**	10.5281/zenodo.16740439
017	2025	Paper	**On Time | Zenodo**	10.5281/zenodo.15168381
018	2025	Paper	**QuadMath: An Analytical Review of 4D and Quadray Coordinates | Zenodo**	10.5281/zenodo.16887800
019	2025	Paper	**ResNei: Solution Design Document | Zenodo**	10.5281/zenodo.15389682
020	2025	Paper	**Symergetics: Symbolic Synergetics for Rational Arithmetic | Zenodo**	10.5281/zenodo.17114389
021	2025	Paper	**Synthesis of Agent and Niche | Zenodo**	10.5281/zenodo.17235137
022	2025	Presentation	**Systems Processes, Active Inference, and Beyond | Presentation**	10.5281/zenodo.17138223
023	2025	Paper	**Temporal Depth in a Coherent Self and in Depersonalization | Frontiers in Psychology**	10.3389/fpsyg.2025.1585315
024	2025	Paper	**The Active Inference Institute & Active Inference Ecosystem (v3, 2025 snapshot) | Zenodo**	10.5281/zenodo.17982447
025	2025	Paper	**The Ant Stack | Zenodo**	10.5281/zenodo.16782756
026	2025	Paper	**The Discovery Engine: AI-Driven Synthesis and Navigation of Scientific Knowledge Landscapes | ArXiv**	10.48550/arXiv.2505.17500
027	2025	Paper	**Thoughtseeds: A Hierarchical and Agentic Framework for Investigating Thought Dynamics in Meditative States | Entropy**	10.3390/e27050459
028	2025	Paper	**Towards a Science of Consciousness and Social Complexity... For Ants | Book Chapter**	https://www.editorafi.org/ebook/c128-colonias-formigas-conscientes
029	2024	Paper	**Aligning Active Inference Ontology to SUMO | Zenodo**	10.5281/zenodo.11459322

#	Year	Type	Title and Venue	Link
030	2024	Presentation	**BioFirm Development at Applied Active Inference Symposium 2024 | Presentation**	10.5281/zenodo.14861595
031	2024	Paper	**Bridging gaps in image meme research: A multidisciplinary paradigm | JASIST**	10.1002/asi.24900
032	2024	Paper	**Chemical and transcriptomic diversity do not correlate with ascending levels of social complexity in Blattodea | Ecology & Evolution**	10.1002/ece3.70063
033	2024	Paper	**Comments on National Digital Twins R&D Strategic Plan | Zenodo**	10.5281/zenodo.13273681
034	2024	Paper	**Enhancing Population-based Search with Active Inference | ArXiv**	10.48550/arXiv.2408.09548
035	2024	Paper	**FarmWorks: Decentralized AI Agents for Personalized Solutions | Zenodo**	10.5281/zenodo.13754585
036	2024	Paper	**Federated inference and belief sharing | Neuroscience & Biobehavioral Reviews**	10.1016/j.neubiorev.2023.105500
037	2024	Paper	**Four-fold Fields of Quantum Dreams | Zenodo**	10.5281/zenodo.10798145
038	2024	Presentation	**MathArt Stream #8: William Blake and Active Inference | Presentation**	10.5281/zenodo.13711301
039	2024	Presentation	**Sensemaking Federation: Exploring the Frontiers of Digital Innovation | Presentation**	10.5281/zenodo.14574046
040	2024	Paper	**Shared Protentions in Multi-Agent Active Inference | Entropy**	10.3390/e26040303
041	2024	Paper	**The Active Inference Institute & Active Inference Ecosystem (v2, 2024 snapshot) | Zenodo**	10.5281/zenodo.14108992
042	2024	Paper	**Way Finding in the Infinite Imaginarium | Zenodo**	10.5281/zenodo.10601082
043	2024	Paper	**Why Paleolithic Rockstars were both enigmatic and sporadic | Physics of Life Reviews**	10.1016/j.plrev.2024.04.010
044	2024	Paper	**Writing on Curio Cards for the "On NFT" book | Taschen**	https://www.worldcat.org/isbn/978-3-8365-9970-2
045	2023	Paper	**A guided tour through the spaces of particular "minds" | Physics of Life Reviews**	10.1016/j.plrev.2023.11.001
046	2023	Paper	**A single-pheromone model accounts for empirical patterns of ant colony foraging | Cognitive Systems Research**	10.1016/j.cogsys.2023.02.005
047	2023	Paper	**A snapshot and pipeline for tissue-specific gene expression meta-analysis in honey bees | Zenodo**	10.5281/zenodo.10400744
048	2023	Paper	**A variational synthesis of evolutionary and developmental dynamics | Entropy**	10.3390/e25070964
049	2023	Paper	**ATLAS: A Question Oriented Approach to Pattern Languages in Knowledge Management | Zenodo**	10.5281/zenodo.10296601
050	2023	Paper	**An Account of Active Inference Modeling | Zenodo**	10.5281/zenodo.8415313
051	2023	Paper	**Cognitive Sovereignty & Active Inference in the State of Exception | Zenodo**	10.5281/zenodo.10038232
052	2023	Paper	**Comments on AI Accountability Policy to NTIA | NTIA**	10.5281/zenodo.8025957
053	2023	Paper	**Distributed Science — The Scientific Process as Multi-Scale Active Inference | OSF**	10.31219/osf.io/dnw5k
054	2023	Paper	**Enhanced NSF Postdoctoral Reporting via Synthetic Intelligence Language Processing | Zenodo**	10.5281/zenodo.10160656
055	2023	Paper	**Experimental Entomology in the Age of Video | JoVE**	10.3791/65002
056	2023	Paper	**Generalized Notation Notation for Active Inference Models | Zenodo**	10.5281/zenodo.7803328
057	2023	Paper	**Generative Research Teams: Active Inference Compositions for Research and Meta-Science | Zenodo**	10.5281/zenodo.8164667
058	2023	Presentation	**Of Ants & Aging | Presentation**	10.5281/zenodo.7855582
059	2023	Paper	**Open Access science needs Open Science Sensemaking (OSSm) | MetaArXiv**	10.31222/osf.io/9nb3u
060	2023	Presentation	**Postdoc review (2020–2023) | Presentation**	10.5281/zenodo.8377988
061	2023	Paper	**The Active Inference Institute and Active Inference Ecosystem (v1) | Zenodo**	10.5281/zenodo.8266281

#	Year	Type	Title and Venue	Link
062	2023	Paper	**The P3IF: Properties, Processes, and Perspectives Inter-Framework | Zenodo**	10.5281/zenodo.10034512
063	2023	Paper	**To comment or not to comment | Physics of Life Reviews**	10.1016/j.plrev.2023.06.002
064	2023	Course	**Transcript of Chris Fields, "Physics as Information Processing" course | Zenodo**	10.5281/zenodo.10447642
065	2023	Paper	**William Blake & Buckminster Fuller: Lives in Juxtaposition | Zenodo**	10.5281/zenodo.7514368
066	2022	Paper	**An Active Inference Ontology for Decentralized Science | Zenodo**	10.5281/zenodo.6320574
067	2022	Paper	**Catechism for Towards Active Diffusion | Zenodo**	10.5281/zenodo.7443848
068	2022	Paper	**From Users to (Sense)Makers: Stigmergic Social Annotation | Hypertext '22**	10.48550/arXiv.2205.06345
069	2022	Paper	**On free will or the lack thereof (interview with Robert Sapolsky) | ALIUS Bulletin**	10.5281/zenodo.7394901
070	2022	Paper	**Predictive Processing Interpretation of the Mirror Test | Zenodo**	10.5281/zenodo.7377256
071	2022	Book	**Structuring the Information Commons: Open Standards and Cognitive Security | COGSEC.org**	https://cogsec.org
072	2022	Paper	**Systems Modeling and Cognitive Audits for Hypercert Ecosystems | Zenodo**	10.5281/zenodo.7626769
073	2022	Paper	**The Free Energy Principle & Active Inference: a Systematic Literature Analysis | Zenodo**	10.5281/zenodo.7449368
074	2022	Paper	**TrustFinder: Recommendations for Community-Based Trust Systems | Zenodo**	10.5281/zenodo.7093837
075	2021	Paper	**Active Inferants: An Active Inference Framework for Ant Colony Behavior | Frontiers in Behavioral Neuroscience**	10.3389/fnbeh.2021.647732
076	2021	Paper	**Active Inference in Modeling Conflict | Zenodo**	10.5281/zenodo.5759807
077	2021	Paper	**Digital Rhetorical Ecosystem Analysis: Sensemaking of Digital Memetic Discourse | Zenodo**	10.5281/zenodo.5573947
078	2021	Playbook	**Innovator's Digital Playbook | Web**	https://innovatorsdigitalplaybook.com/
079	2021	Book	**Narrative Information Ecosystems: Conflict and Trust on the Endless Frontier | COGSEC.org**	https://cogsec.org
080	2021	Paper	**Narrative Information Management | Zenodo**	10.5281/zenodo.5573287
081	2021	Playbook	**The Facilitator's Catechism Playbook | Zenodo**	10.5281/zenodo.4579414
082	2020	Paper	**Active Inference & Behavior Engineering for Teams | Zenodo**	10.5281/zenodo.4021163
083	2020	Course	**Communication for Remote Teams | Udemy**	https://www.udemy.com/course/communication-for-remote-teams/
084	2020	Paper	**Distributed physiology and the molecular basis of social life in eusocial insects | Hormones & Behavior**	10.1016/j.yhbeh.2020.104757
085	2020	Paper	**Emergent Teams for Complex Threats | Zenodo**	10.5281/zenodo.3986085
086	2020	Paper	**Gene expression variation in the brains of harvester ant foragers is associated with collective behavior | Communications Biology**	10.1038/s42003-020-0813-8
087	2020	Course	**How to use Keybase for remote teams | Udemy**	https://www.udemy.com/course/keybase-for-remote-teams/
088	2020	Paper	**Measurement of natural variation of neurotransmitter tissue content in red harvester ant brains | Analytical & Bioanalytical Chemistry**	10.1007/s00216-019-02355-3
089	2020	Book	**The Great Preset: Remote Teams and Operational Art | COGSEC.org**	https://cogsec.org
090	2020	Paper	**The Innovator's Catechism | Zenodo**	10.5281/zenodo.4383230
091	2020	Paper	**The neuroscience of decision making (interview with Timothy Hanks) | ALIUS Bulletin**	10.34700/8pq4-0h12
092	2019	Paper	**Dennett Explained (interview with Daniel Dennett) | ALIUS Bulletin**	10.34700/7qkw-zh08
093	2019	Paper	**PhD: Behavioral, Physiological, and Transcriptomic Variation Among Colonies of Pogonomyrmex barbatus | Stanford University**	http://purl.stanford.edu/pb813wm1484
094	2019	Paper	**The ant colony as a test for scientific theories of consciousness | Synthese**	10.1007/s11229-019-02130-y

#	Year	Type	Title and Venue	Link
095	2019	Paper	**The physiology of forager hydration and variation among harvester ant colonies | Scientific Reports**	10.1038/s41598-019-41586-3
096	2018	Paper	**Foraging behavior and locomotion of the invasive Argentine ant from winter aggregations | PLoS One**	10.1371/journal.pone.0202117
097	2018	Presentation	**MVEE: A Framework for Evolutionary Studies | Presentation**	10.5281/zenodo.13999298
098	2018	Paper	**Of woodlice and men: A Bayesian account of cognition, life and consciousness (with Karl Friston) | ALIUS Bulletin**	10.34700/h460-nz89
099	2018	Paper	**Partner Pen Play in Parallel (PPPiP): A New Paradigm for Relationship Improvement | Arts**	10.3390/arts7030039
100	2018	Paper	**The Role of Dopamine in Collective Regulation of Foraging in Harvester Ants | iScience**	10.1016/j.isci.2018.09.001
101	2017	Paper	**The MutAnts are here | Cell**	10.1016/j.cell.2017.07.046
102	2017	Paper	**Two lineages that need each other | Molecular Ecology**	10.1111/mec.13964
103	2016	Paper	**Ant Genetics: Reproductive Physiology, Worker Morphology, and Behavior | Annual Review of Neuroscience**	10.1146/annurev-neuro-070815-013927
104	2016	Paper	**Context-dependent expression of the foraging gene in field colonies of ants | Proceedings of the Royal Society B**	10.1098/rspb.2016.0841
105	2016	Paper	**Influence of Nuclear Structure on formation of radiation-induced lethal lesions | Int. J. Radiation Biology**	10.3109/09553002.2016.1144941
106	2015	Paper	**Commentary: Portuguese crypto-Jews: the genetic heritage of a complex history | Frontiers in Genetics**	10.3389/fgene.2015.00261
107	2015	Paper	**Could Ehrlichial Infection Cause Some Changes Associated with Leukemia? | Medical Hypotheses**	10.1016/j.mehy.2015.09.015
108	2015	Paper	**Large Scale Coding Sequence Change Underlies the Evolution of Post-developmental Novelty in Honey Bees | Molecular Biology & Evolution**	10.1093/molbev/msu292
109	2026	Paper	**Ento-Linguistics: Language, Ambiguity, and Scientific Communication in Entomology | Zenodo**	10.5281/zenodo.19574118
110	2026	Paper	**A Living Meta-Analysis Architecture for Active Inference: Assertion Extraction, Nanopublications, and Hypothesis Scoring | Active Inference Journal**	10.5281/zenodo.19897664
111	2026	Paper	**Dynamic Attentional Agents in Focused Attention Meditation: Hierarchical Computational Modeling of Expert-Novice Differences | CSCIS vol 2857, Springer**	10.1007/978-3-032-16955-6_11
112	2026	Paper	**Compositional Approaches to Linguistic Case for Cognitive Modeling | Active Inference Journal**	10.5281/zenodo.19695260
113	2026	Paper	**Towards Lean 4 Formalization of the Free Energy Principle: AI-Driven Theorem Sketching and Verification for Active Inference and Bayesian Mechanics | Active Inference Journal**	10.5281/zenodo.19699234
114	2020	Paper	**The Facilitator's Catechism | Zenodo**	10.5281/zenodo.4203765
115	2026	Paper	**The Architecture of False Gods: William Blake, Professor Jiang, and the Active Inference Corrective to Single Vision | Zenodo**	10.5281/zenodo.20144984
116	2026	Paper	**Crescent City in Living Waves: Space, Time, People, and Minds on the Southern Cascadian Coast | Zenodo**	10.5281/zenodo.20286171
117	2026	Book	**Introduction to Biology: A Generative Approach | Zenodo**	10.5281/zenodo.20286478
118	2025	Paper	**CEREBRUM: Case-Enabled Reasoning Engine with Bayesian Representations for Unified Modeling | Zenodo**	10.5281/zenodo.15231156
119	2026	Paper	**Policy Entanglement in Active Inference | Zenodo**	10.5281/zenodo.20419637
120	2026	Paper	**Bounded AutoResearch for a Tiny Reproducible Machine-Learning Task | Zenodo**	10.5281/zenodo.20420357
121	2026	Paper	**Convergence Analysis of Gradient Descent Optimization | Zenodo**	10.5281/zenodo.20420368
122	2026	Paper	**Editorial Quality at Scale: A Reproducible Prose-Review Pipeline | Zenodo**	10.5281/zenodo.20420342
123	2026	Paper	**A template/ approach to Reproducible Generative Research | Zenodo**	10.5281/zenodo.20420387

#	Year	Type	Title and Venue	Link
124	2026	Paper	**Active Inference Multi-Track Exemplar | Zenodo**	10.5281/zenodo.20420352
125	2026	Paper	**BeeStack: An Evidence-Typed Scaffold for Whole-Colony Honeybee Simulation | Zenodo**	10.5281/zenodo.20420557
126	2026	Paper	**When do bugs see (infra)red? | Zenodo**	10.5281/zenodo.20450970
127	2026	Paper	**Self-Improvement Agent Harness: A Deterministic SIA Exemplar | Zenodo**	10.5281/zenodo.20453880
128	2025	Presentation	**Biofirm Development with First Principles First and the Active Inference Institute at the Applied Active Inference Symposium 2024 | Zenodo**	10.5281/zenodo.14861596
129	2025	Paper	**Con-cat-enate: emulation of Cat Hippocampus | Zenodo**	10.5281/zenodo.14738798
130	2025	Paper	**Graphspeak: of language and handshake | Zenodo**	10.5281/zenodo.14737157
131	2025	Paper	**BevCyc - posit on primitive drivers of creatures | Zenodo**	10.5281/zenodo.14737076
132	2025	Paper	**Beacons - pre emulation of social cortex | Zenodo**	10.5281/zenodo.14737060
133	2025	Paper	**Con-cat-enate: pilot overview | Zenodo**	10.5281/zenodo.14737043
134	2024	Paper	**Push and Pull: A priming sequence | Zenodo**	10.5281/zenodo.10659375
135	2023	Paper	**A Natural AI Based on The Science of Computational Physics, Biology and Neuroscience: Policy and Societal Significance | Zenodo**	10.5281/zenodo.10360148
136	2023	Paper	**ATLAS: A Question Oriented Approach to the Use of Pattern Languages in Knowledge Management | Zenodo**	10.5281/zenodo.10362561
137	2023	Paper	**Transcript of Active Inference GuestStream 049.1: "Clickbait, consciousness science, and responsible journalism" | Zenodo**	10.5281/zenodo.8229512
138	2023	Paper	**Modern Nostr Index Card-based Knowledge Engineering | Zenodo**	10.5281/zenodo.8118156
139	2023	Paper	**Slides for Iris | Zenodo**	10.5281/zenodo.7838653
140	2022	Paper	**Canonical neural networks perform active inference | Zenodo**	10.5281/zenodo.7400536
141	2022	Paper	**Interoception as modeling, allostasis as control | Zenodo**	10.5281/zenodo.7400709
142	2022	Paper	**Transcript of: Mark Solms, "Consciousness as Precision Optimization: Some Physiological and Philosophical Considerations", ActInf GuestStream #016 | Zenodo**	10.5281/zenodo.7267947
143	2022	Paper	**A Worked Example of the Bayesian Mechanics of Classical Objects | Zenodo**	10.5281/zenodo.7400786
144	2022	Paper	**Transcript of discussions on: "Communication as Socially Extended Active Inference: An Ecological Approach to Communicative Behavior" | Zenodo**	10.5281/zenodo.7401875
145	2022	Paper	**Tracking Public Sensemaking through Rhetorical Annotation of Image Memes | Zenodo**	10.5281/zenodo.6904427
146	2021	Paper	**Transcript of: Karl Friston, 1st Applied Active Inference Symposium, Active Inference Lab, June 21, 2021 | Zenodo**	10.5281/zenodo.5797072
147	2021	Paper	**Collaborative Writing for Catechism-Based Teams | Zenodo**	10.5281/zenodo.4633921
148	2020	Paper	**Reimagining Maps | Zenodo**	10.5281/zenodo.4170026
149	2020	Paper	**Infinite Games for Infinite Teams | Zenodo**	10.5281/zenodo.12601675
150	2026	Paper	**The Music Never Stopped: A Grateful Data Compendium with a Category-Theoretic Interpretation | Zenodo**	10.5281/zenodo.20482026
151	2026	Paper	**A Deterministic Testbed for Self-Organizing Agent-Team Coordination | Zenodo**	10.5281/zenodo.20533670
152	2026	Paper	**Recovering LLM-Persona Accuracies from Unlabeled Votes | Zenodo**	10.5281/zenodo.20498700
153	2026	Paper	**The Triplicate: A Data-Driven Large-Format Newspaper Layout Engine | Zenodo**	10.5281/zenodo.20533676
154	2026	Book	**The Template Textbook | Zenodo**	10.5281/zenodo.20533126

Software (88)

Repository	Stack	Description and Source Links
docxology/template https://github.com/docxology/template	Python	Production-grade scaffold for reproducible computational research — 10-stage DAG pipelines, ≥90% test coverage, cryptographic provenance, multi-project workspace, and agent-ready documentation · Zenodo software v3.3.0 zenodo: https://doi.org/10.5281/zenodo.20584820
docxology/QuadCraft https://github.com/docxology/QuadCraft	JavaScript	Minecraft variant built on tetrahedral geometry — voxels replaced by Quadray-coordinate tetrahedra, exploring Buckminster Fuller's synergetics in an interactive 3D world
docxology/MetaInformAnt https://github.com/docxology/MetaInformAnt	Python	Meta-framework integrating computational entomology, Active Inference, and information theory for modeling ant colony cognition and beyond paper: papers/2021_ActiveInferants/
docxology/codomyrmex https://github.com/docxology/codomyrmex	Python	AI-native modular coding workspace — 128 modules, 600 MCP tools, 21K+ zero-mock tests, multi-agent orchestration (Claude/Gemini/GPT), PAI integration
docxology/cognitive-engine https://github.com/docxology/cognitive-engine	Python	Hybrid neuro-symbolic AI combining fractal symbolic representations with probabilistic neural models for enhanced reasoning; includes Unipixel fractal building blocks and PEFF ethical reasoning
docxology/opentir https://github.com/docxology/opentir	Python	Comprehensive toolkit analyzing 250+ Palantir open-source repositories (6.2M+ lines of code) — automated discovery, multi-language analysis, dependency mapping, statistical profiling, and professional visualizations
docxology/timeline_generator https://github.com/docxology/timeline_generator	TypeScript	Networked Life Encoding — interactive D3 knowledge graph of human intellectual history; 24 relationship types, Perplexity AI enrichment; seeded around R. Buckminster Fuller's network
docxology/FORMINDEX https://github.com/docxology/FORMINDEX	Rich Text Format	Analysis and tooling around FORMIS, the world's largest ant-literature database; bibliometric mining and structured index generation
docxology/ActiveInferAnts https://github.com/docxology/ActiveInferAnts	Python	Active Inference for ants — multi-language implementations (Python) for modeling ant-colony cognition as Bayesian agents minimizing free energy paper: papers/2021_ActiveInferants/
docxology/crescent-city https://github.com/docxology/crescent-city	TypeScript	End-to-end pipeline for ingesting, parsing, and querying the Crescent City, CA municipal code; TypeScript-based legal-text processing
docxology/ivm-xyz https://github.com/docxology/ivm-xyz	Python	Python library for 3D and 4D Quadray/IVM (Isotropic Vector Matrix) geometry — coordinate conversion, volume calculations, and polyhedral operations
docxology/Markov_Blanket_Detection https://github.com/docxology/Markov_Blanket_Detection	Python	Dynamic Markov Blanket Detection (DMBD) — PyTorch-accelerated framework for identifying causal relationships in complex systems; static and temporal modes with cognitive-structure labeling
docxology/QuadMath https://github.com/docxology/QuadMath	TeX	Mathematical exposition of Quadray coordinate systems (4-vector tetrahedral coordinates); LaTeX documentation and computation library
docxology/active_inference https://github.com/docxology/active_inference	Python	Active Inference for, with, and by Generative AI — Python implementations of free-energy-minimizing agents integrated with modern AI tooling paper: papers/2020_BehaviorEngineering/
docxology/AgenticMesh https://github.com/docxology/AgenticMesh	Python	Active Inference & Agentic Mesh — modular Python infrastructure for building agentic systems using free-energy-minimizing Active Inference principles
docxology/curriculum https://github.com/docxology/curriculum	HTML	Curriculum development framework for interdisciplinary education in Active Inference, biology, and computational science
docxology/lean_niche https://github.com/docxology/lean_niche	Python	Lean/minimal niche-construction modeling — formal computational representations of ecological niche dynamics
docxology/literature https://github.com/docxology/literature	Python	Literature collection and analysis tools for structured ingestion, annotation, and exploration of scientific publications
docxology/obsidian-construction-from-text https://github.com/docxology/obsidian-construction-from-text	Python	Automated Obsidian vault construction from unstructured text — NLP pipelines to generate bidirectionally-linked knowledge graphs
docxology/qr_live_protocol https://github.com/docxology/qr_live_protocol	Python	QR code-based live protocol system for real-time data exchange and interactive session management

Repository	Stack	Description and Source Links
docxology/template_autoscientists https://github.com/docxology/template_autoscientists	Python	AutoScientists deterministic multi-agent scientific-discovery coordination harness · · Zenodo paper: papers/2026_DeterministicTestbedSelf/ | zenodo: https://doi.org/10.5281/zenodo.20533670
docxology/template_newspaper https://github.com/docxology/template_newspaper	Python	The Triplicate data-driven large-format newspaper layout engine · · Zenodo paper: papers/2026_Triplicate/ | zenodo: https://doi.org/10.5281/zenodo.20533676
docxology/ntqr_llm https://github.com/docxology/ntqr_llm	Python	Algebraic NTQR evaluation study recovering LLM-persona accuracies from unlabeled votes · · Zenodo paper: papers/2026_RecoveringLLMPersona/ | zenodo: https://doi.org/10.5281/zenodo.20498700
docxology/ultralink-docx https://github.com/docxology/ultralink-docx	HTML	UltraLink document format tooling — linking, rendering, and transformation utilities for rich hyperlinked documents
docxology/cognitive https://github.com/docxology/cognitive	Python	Cognitive Ecosystem Modeling Framework — Active Inference agents with Obsidian-compatible knowledge management, bidirectional graph validation, belief updating, and network visualization
docxology/active_torchference https://github.com/docxology/active_torchference	Python	PyTorch-based Active Inference implementations — GPU-accelerated variational inference and free energy minimization
docxology/ant-pheromone https://github.com/docxology/ant-pheromone	Python	Computational simulation of ant pheromone trail dynamics and stigmergic communication
docxology/ant_stack https://github.com/docxology/ant_stack	Python	The Ant Stack — layered computational model of collective ant-colony intelligence paper: papers/2025_AntStack/
docxology/cohereants https://github.com/docxology/cohereants	Python	Computational research on insect infrared sensing and olfaction modeling · · Zenodo paper: papers/2026_WhenDoBugs/ | zenodo: https://doi.org/10.5281/zenodo.20450970
docxology/ento_linguistics https://github.com/docxology/ento_linguistics	Python	Ento-Linguistics corpus pipeline — term extraction, terminology networks, semantic entropy, CACE scoring paper: papers/2026_EntoLinguistics/
docxology/cognitive_case_diagrams https://github.com/docxology/cognitive_case_diagrams	TeX	Compositional linguistic case for cognitive modeling — categorical case systems, DisCoCat/DisCoCirc diagrams, tests and figures for the Active Inference Journal manuscript paper: papers/2026_CognitiveCaseDiagrams/
docxology/Autonomous-Drone-Navigation-in-AirSim-With-Active-Inference https://github.com/docxology/Autonomous-Drone-Navigation-in-AirSim-With-Active-Inference	Julia	Active Inference-driven autonomous drone navigation in Microsoft AirSim simulation environment
docxology/biol-1 https://github.com/docxology/biol-1	HTML	BIOL-1: General Biology course materials — College of the Redwoods; lectures, labs, and interactive HTML content
docxology/biol-8 https://github.com/docxology/biol-8	HTML	BIOL-8: Human Biology course materials — College of the Redwoods; lectures, labs, and interactive HTML content
docxology/biology_textbook https://github.com/docxology/biology_textbook	Python	Open generative biology textbook — Markdown source, tested Python modules, programmatic figures; archived at Zenodo paper: papers/2026_BiologyTextbook/ | zenodo: https://doi.org/10.5281/zenodo.20286478
docxology/cascadia https://github.com/docxology/cascadia	Python	Cascadia bioregion modeling and data analysis tools
docxology/course https://github.com/docxology/course	Python	Course materials framework for structured interdisciplinary learning modules
docxology/enactive_inference_model https://github.com/docxology/enactive_inference_model	Python	Three-level hierarchical enactive inference model of mental action — focused-attention meditation with expert/novice profiles; reproducible simulations and figures paper: papers/2025_Thoughtseeds/
docxology/EvoJump https://github.com/docxology/EvoJump	Python	Comprehensive Framework for Evolutionary Ontogenetic Analysis — modeling developmental trajectories and evolutionary dynamics paper: papers/2025_EvoJump/
docxology/flick https://github.com/docxology/flick	Python	Lightweight Python tooling for rapid data flicking and processing pipelines

Repository	Stack	Description and Source Links
docxology/fuller-obsidian https://github.com/docxology/fuller-obsidian	unspecified	Obsidian vault dedicated to R. Buckminster Fuller's ideas — synergetics, geodesics, and design science
docxology/godel_ivm https://github.com/docxology/godel_ivm	Python	Gödel numbering meets the IVM (Isotropic Vector Matrix) — formal encoding of geometric objects as prime products
docxology/goference https://github.com/docxology/goference	Go	Active Goference — production-ready Go framework for Active Inference autonomous agents; genuine variational inference, free energy minimization, POMDP policy planning via gonum
docxology/hhs-opendata https://github.com/docxology/hhs-opendata	Python	Analysis of HHS (Department of Health and Human Services) open data — policy analytics and public-health data pipelines
docxology/grateful_data https://github.com/docxology/grateful_data	Python	The Music Never Stopped citation-bound Grateful Dead data compendium · · Zenodo paper: papers/2026_MusicNeverStopped/ | zenodo: https://doi.org/10.5281/zenodo.20482026
docxology/infra-calc https://github.com/docxology/infra-calc	JavaScript	Infrastructure calculator for estimating resource requirements and costs in multi-agent and AI systems
docxology/markdown_decision_process https://github.com/docxology/markdown_decision_process	Python	Framework for Probabilistic Document Analysis — Markov Decision Process over Markdown artifacts for structured document reasoning paper: papers/2025_MarkdownDecisionProcess/
docxology/mdkv https://github.com/docxology/mdkv	Python	MKV-like key-value format and methods for Markdown — structured metadata embedding in plain-text documents paper: papers/2025_MDKV/
docxology/p3if https://github.com/docxology/p3if	Python	Properties, Processes, and Perspectives Inter-Framework (P3IF) — multiplexing interdisciplinary requirements frameworks to manage information risk and foster cognitive security paper: papers/2023_P3IF/
docxology/service https://github.com/docxology/service	Python	Service-layer infrastructure and API tooling for Python-based research applications
docxology/snake https://github.com/docxology/snake	Python	Classic snake game implemented in Python — used for testing multi-agent and reinforcement learning algorithms
docxology/steganographer https://github.com/docxology/steganographer	Rust	High-performance Rust tool for embedding BLAKE3+Ed25519 cryptographic signatures and visible watermarks into live video/audio via LSB steganography; four Rust crates, 132 tests, GStreamer integration
docxology/template_textbook https://github.com/docxology/template_textbook	HTML	Modular fillable scaffold for book-length technical works · · Zenodo paper: papers/2026_TemplateTextbook/ | zenodo: https://doi.org/10.5281/zenodo.20533126
docxology/blake_jiang https://github.com/docxology/blake_jiang	Python	William Blake, Professor Jiang, and the Architecture of Intelligence — AI-augmented scholarly response to Game Theory #24 with Blake, Jiang, and Active Inference framing paper: papers/2026_BlakeJiang/
docxology/symergetics https://github.com/docxology/symergetics	Python	Symbolic Synergetics — computer algebra and formal symbolic systems for Buckminster Fuller's synergetics mathematics
docxology/transformer https://github.com/docxology/transformer	Python	Transformer architecture explorations — attention mechanisms and architectural variants for Active Inference research
ActiveInferenceInstitute/ActiveInferenceJournal https://github.com/ActiveInferenceInstitute/ActiveInferenceJournal	HTML	Content hub for the Active Inference Journal — 500+ video transcripts, stream archives, and multimedia publishing infrastructure for the Institute's primary educational output
ActiveInferenceInstitute/ActiveBlockference https://github.com/ActiveInferenceInstitute/ActiveBlockference	Jupyter	Active Inference agents in cadCAD block-based simulation — formal agent-based modeling using complex-systems simulation framework
ActiveInferenceInstitute/ActiveInferAnts https://github.com/ActiveInferenceInstitute/ActiveInferAnts	Python	Active Inference models for/of ants across 32+ programming languages — multi-language reference implementation of stigmergic Active Inference
ActiveInferenceInstitute/GeneralizedNotationNotation https://github.com/ActiveInferenceInstitute/GeneralizedNotationNotation	Python	GNN — formal notation standard for specifying Active Inference generative models; JSON/YAML schema, validators, and transpilers to Python, Julia, and MATLAB · Zenodo software zenodo: https://doi.org/10.5281/zenodo.19600217

Repository	Stack	Description and Source Links
ActiveInferenceInstitute/fep_lean https://github.com/ActiveInferenceInstitute/fep_lean	Python	Lean 4 / Mathlib4 formalization of FEP-related mathematics — 50-topic sorry-free catalog, Hermes/OpenGauss LLM drafting, zero-mock verification, manuscript pipeline · · Zenodo paper: papers/2026_FEPLean/ | zenodo: https://doi.org/10.5281/zenodo.19699234
ActiveInferenceInstitute/cognitive https://github.com/ActiveInferenceInstitute/cognitive	Python	Cognitive modeling and simulation tools — multi-agent Active Inference environments, benchmarks, and visualization
ActiveInferenceInstitute/Active_Inference_Ontology https://github.com/ActiveInferenceInstitute/Active_Inference_Ontology	unspecified	Formal OWL/RDF ontology for Active Inference — defines classes, properties, and axioms for free energy principle concepts
ActiveInferenceInstitute/Journal-Utilities https://github.com/ActiveInferenceInstitute/Journal-Utilities	HTML	Transcription, processing, and publishing pipelines for the Active Inference Journal — Whisper-based download, transcription, RAG/knowledge-graph, and export utilities · Zenodo release zenodo: https://doi.org/10.5281/zenodo.18686966
ActiveInferenceInstitute/Symposium https://github.com/ActiveInferenceInstitute/Symposium	Python	Applied Active Inference Symposium materials — talks, workshops, proceedings, and collaborative session artifacts from annual Applied AIF symposia
ActiveInferenceInstitute/ActiveInferenceCategoryTheory https://github.com/ActiveInferenceInstitute/ActiveInferenceCategoryTheory	unspecified	Formal categorical foundations for Active Inference — string diagrams, functorial semantics, and compositional models using category theory
ActiveInferenceInstitute/CEREBRUM https://github.com/ActiveInferenceInstitute/CEREBRUM	Python	Case-Enabled Reasoning Engine with Bayesian Representations for Unified Modeling — grammatical case framework for compositional cognitive models · Zenodo paper: papers/2025_CEREBRUM2/ | zenodo: https://doi.org/10.5281/zenodo.15231156
ActiveInferenceInstitute/GEO-INFER https://github.com/ActiveInferenceInstitute/GEO-INFER	HTML	Geospatial Active Inference — applying free energy minimization to geographic systems, spatial planning, and bioregional intelligence
ActiveInferenceInstitute/AEOS https://github.com/ActiveInferenceInstitute/AEOS	unspecified	Active Entity Ontology for Science — OWL ontology for representing active entities, their actions, and causal relationships in scientific models
ActiveInferenceInstitute/courses https://github.com/ActiveInferenceInstitute/courses	HTML	Structured Active Inference learning courses — cohort-based curriculum materials, exercises, and collaborative learning artifacts
ActiveInferenceInstitute/Research-Discovery-Engine https://github.com/ActiveInferenceInstitute/Research-Discovery-Engine	HTML	AI-driven research synthesis and navigation — semantic search, citation graph analysis, and cross-domain connection discovery across the Active Inference literature
ActiveInferenceInstitute/Parr_et_al_2022_ActInf_Textbook https://github.com/ActiveInferenceInstitute/Parr_et_al_2022_ActInf_Textbook	Julia	Active Inference Textbook Study Group — code, exercises, and reading notes accompanying Parr, Pezzulo & Friston (2022)
ActiveInferenceInstitute/Biofirm https://github.com/ActiveInferenceInstitute/Biofirm	Python	Bioregional Active Inference for organizational stewardship — applying free energy minimization to ecological and community governance systems
ActiveInferenceInstitute/Start https://github.com/ActiveInferenceInstitute/Start	Python	Institute onboarding hub — getting-started guides, contributor pathways, project index, and orientation materials for new participants
ActiveInferenceInstitute/Knowledge-Engineering https://github.com/ActiveInferenceInstitute/Knowledge-Engineering	Jupyter	Knowledge engineering tools and workflows — ontology alignment, concept mapping, and structured knowledge-base construction for Active Inference
ActiveInferenceInstitute/ActInf_RxInfer https://github.com/ActiveInferenceInstitute/ActInf_RxInfer	unspecified	Active Inference with RxInfer.jl — Julia-based reactive message passing for efficient Bayesian inference in Active Inference models
ActiveInferenceInstitute/pymcp https://github.com/ActiveInferenceInstitute/pymcp	Python	PyMCP — Model Context Protocol bridge for PyMDP, enabling Active Inference agents to interoperate with MCP-compatible AI tooling
ActiveInferenceInstitute/GEN24 https://github.com/ActiveInferenceInstitute/GEN24	Jupyter	GEN-24 working project — collaborative generative systems research from the 2024 active inference cohort

Repository	Stack	Description and Source Links
ActiveInferenceInstitute/NoOrg https://github.com/ActiveInferenceInstitute/NoOrg	HTML	NoOrg is a Real Org — experimental organizational infrastructure and governance frameworks for decentralized scientific institutions
ActiveInferenceInstitute/ultralink https://github.com/ActiveInferenceInstitute/ultralink	unspecified	UltraLink — framework for linking, rendering, and transforming richly cross-referenced documents across multiple output formats
ActiveInferenceInstitute/ActiveInferenceImplementations https://github.com/ActiveInferenceInstitute/ActiveInferenceImplementations	Jupyter	Learning and exploring probabilistic inference — beginner-accessible Jupyter notebooks for Active Inference from first principles
ActiveInferenceInstitute/ATLAS https://github.com/ActiveInferenceInstitute/ATLAS	Python	Question-Oriented Pattern Languages for Knowledge Management — structured templates for navigating complex scientific knowledge spaces
ActiveInferenceInstitute/Bots https://github.com/ActiveInferenceInstitute/Bots	unspecified	Institute bot infrastructure — automation, notification, and community management bots for the Active Inference Institute's platforms
ActiveInferenceInstitute/Garden-of-Iris https://github.com/ActiveInferenceInstitute/Garden-of-Iris	Python	Interactive visualization garden — generative art and data-visualization experiments exploring Active Inference dynamics
ActiveInferenceInstitute/gen25 https://github.com/ActiveInferenceInstitute/gen25	unspecified	GEN-25 working project — generative systems and collaborative research from the 2025 active inference cohort
ActiveInferenceInstitute/gnn https://github.com/ActiveInferenceInstitute/gnn	unspecified	Generalized Notation Notation v2 — next-generation GNN specification with extended model classes and improved toolchain
ActiveInferenceInstitute/ISSS https://github.com/ActiveInferenceInstitute/ISSS	Python	International Society for the Systems Sciences collaboration — joint research and educational materials bridging Active Inference and systems science
ActiveInferenceInstitute/topp https://github.com/ActiveInferenceInstitute/topp	TeX	The Open Problems Project — curated registry of open scientific problems in Active Inference and the free energy principle

Conferences, Posters, Seminars, Workshops (17)

2025 **Inference about Foraging and Foraging as Inference | Queens College, CUNY, Biology**

- Talk

Source: Queens College Biology https://www.qc.cuny.edu/academics/bio/

2025 **Systems Processes, Active Inference, and Beyond | Enduring Patterns, Emerging Futures: Celebrating Dr. Len Troncale**

- Talk 2025-09-17

Source: Zenodo presentation https://doi.org/10.5281/zenodo.17138223

2024 **...And Eternity in an hour of Systems Science Education and Knowledge Engineering | ISSS 2024**

- Workshop

Source: ISSS https://isss.org/world/

2024 **Funding AI public goods and keeping them public | Funding the Commons**

- Panel

Source: Funding the Commons https://fundingthecommons.io/

2024 **Intelligence in the Age of LLMs | European Identity and Cloud Conference 2024**

- Panel

Source: KuppingerCole EIC speaker https://www.kuppingercole.com/events/eic2024/speakers/3516

2022 **Tissue-specific gene expression meta-analysis in honey bees (Apis mellifera) | Entomology Society of America, Pacific Branch**

- Talk 2022-04-10

Source: Entomological Society of America https://entsoc.org/events

2021 **Thinking like a State: Active inference and the deep roots of complex societies | Cognitio 2021**

- Talk Avel Guenin-Carlut, Daniel Friedman, and Virginia Bleu Knight.

Source: Cognitio https://cognitio.uqam.ca/

2021 **Introducing Active Inference as An Integrative Principle in The Study of Collective Intelligence | ACM-CI 2021**

- Talk Avel Guenin-Carlut and Daniel Friedman.

Source: ACM Collective Intelligence https://ci.acm.org/

2020 **Active Inference for Communicating Systems | IIBM, Chile**

- Talk Invited online seminar. 2020-11-03

Source: IIBM https://www.iibm.cl/

2019 **Invited seminar, Department of Animal Behavior, University of California, Davis | Essig Brunch Seminar Series**

- Talk 2019-04-05

Source: UC Davis Entomology https://entomology.ucdavis.edu/

2018 **Invited seminar, SUMS-RAS | Stanford University Mass Spectrometry Research Symposium**

- Talk 2018-10-04

Source: Stanford Mass Spectrometry https://mass-spec.stanford.edu/

2018 **Invited seminar, University of California, Berkeley | Essig Entomology Brunch**

- Talk 2018-02-06

Source: Essig Museum https://essig.berkeley.edu/

2017 **The role of dopamine in the regulation of foraging in the red harvester ant | Entomology 2017**

- Poster

Source: Entomological Society of America https://entsoc.org/events/annual-meeting

2017 **Quantitative LC-MS/MS analysis of dopamine in single ant brain samples | ASMS 2017**

- Poster KM Krasinska, DA Friedman, DM Gordon, AS Chien.

Source: ASMS Annual Conference https://www.asms.org/conferences/annual-conference

2015 **Transcriptomic analysis of the regulation of foraging in harvester ants | Biology and Genetics of Social Insects, CSHL 2015**

- Poster DA Friedman, A Pilko, DM Gordon.

Source: Cold Spring Harbor Meetings https://meetings.cshl.edu/

2014 **Evolution of sex comb enhancers in the HOX gene Scr | Drosophila Research Conference 2014**

- Poster DA Friedman, T Orenic, C McCullough, E Eksi, OY Barmina, A Kopp.

Source: GSA Drosophila Conference https://genetics-gsa.org/drosophila/

2012 **Visualizing the Integration of Sensory Neurons into a Sex-Specific Central Nervous System Circuit in 3-D | UC Davis Undergraduate Research Conference**

- Talk DA Friedman.

Source: UC Davis Undergraduate Research Center https://urc.ucdavis.edu/

Science Engagement, Outreach, Quotes, Articles (29)

2026 **S3E10 The FEP: A Tool for Understanding and Modelling Complex Systems**

- Video conversation on another channel

Source: BathmikeC podcast https://bathmikec.podbean.com/e/s3e10-the-fep-a-tool-for-understanding-and-modelling-complex-systems/

2025 **(DIA 1/SESSAO 1) Lancamento do Livro: As Colonias de Formigas Sao Conscientes?**

- Video conversation on another channel

Source: YouTube https://www.youtube.com/watch?v=BI-PAt3cgf4

2024 **From Ants to Active Inference with Daniel Ari Friedman**

- Video conversation on another channel | Love and Philosophy, Andrea Hiott

Source: YouTube https://www.youtube.com/watch?v=x_VPw1K55BM

2024 **Building Smarter AI: Active Inference and Nested Models**

- Video conversation on another channel | Bagaev, de Vries, Friedman, Pashea.

Source: YouTube https://youtu.be/u3w24FtAsKw

2024 **Interaction. In Search of Daniel Friedman's Deepest Value**

- Video conversation on another channel | Math4Wisdom

Source: YouTube https://www.youtube.com/watch?v=T05uLRbcloE

2023 **Active Inference and The Free Energy Principle with Daniel Friedman**

- Video conversation on another channel | No Way Out podcast

Source: YouTube https://www.youtube.com/watch?v=pYleeDb-B5E

2023 **Episode 6: Daniel Friedman - Active Inference Institute - What is Active Inference?**

- Video conversation on another channel | Denise Holt

Source: YouTube https://www.youtube.com/watch?v=ZyFqRBp1KE0

2022 **Artist of Curio Cards 24-26 Shares Incredible Journey through Art and Science**

- Video conversation on another channel | Jake Gallen

Source: YouTube https://youtu.be/X_F2YFIiUnY

2022 **What Ant Behavior Reveals About Complexity Science with Daniel Friedman**

- Video conversation on another channel | Making Sense of Complexity

Source: YouTube https://youtu.be/lOr98-sG8r0 | YouTube episode https://www.youtube.com/watch?v=4gsyV38gfTw

2022 **What is Active Inference?...with Daniel Friedman**

- Video conversation on another channel | Inner Sense

Source: YouTube https://youtu.be/yHnm-zMqLvY

2022 **Great discussion with Daniel, OG Curio Card artist, cards 24, 25, 26**

- Video conversation on another channel | ZeroG

Source: YouTube https://youtu.be/AHOGPsqIWmU

2022 **Daniel Friedman Interview - Adventures in NFTs #3**

- Video conversation on another channel | World Crypto Network

Source: YouTube https://www.youtube.com/watch?v=mU_IwFsEpNg

2022 **Banging the Drum with Daniel Friedman**

- Video conversation on another channel | Nicholas Pacheco

Source: YouTube https://www.youtube.com/watch?v=fBuS4vItitw

2022 **KB7 Fireside: Scaling Principled Games**

- Video conversation on another channel | Kernel

Source: YouTube https://www.youtube.com/watch?v=vFOk8UR-UVQ

2022 **KBCast Episode 7: Kernel Block 5 Week 6 and special guest Daniel Ari Friedman**

- Video conversation on another channel | Active Inference Lab

Source: YouTube https://www.youtube.com/watch?v=R5zdTez8CPk

2021 **Daniel Friedman Interview - Adventures in NFTs #3**

- Video conversation on another channel | World Crypto Network

Source: YouTube https://youtu.be/mU_IwFsEpNg

2020 **Interview with Monica Kang about Complexity Weekend, with co-founder Shaun Swanson and organizer JP Gonzales**

- Video conversation on another channel

Source: YouTube https://youtu.be/nEaG4xj6MCE

2021 **Blurb in Science magazine inventing the word disinforge**

- Blurb

Source: Science https://www.science.org/doi/10.1126/science.abg0904

2021 **Complexity Weekend featured on InnovatorsBox blog, with quotes from Friedman and others**

- Blurb

Source: InnovatorsBox https://blog.innovatorsbox.com/complexity-weekend/

2020 **Blurb in Science magazine about rethinking time use for scientists**

- Blurb

Source: Science https://www.science.org/

2016 **Blurb in Science magazine about what ants can teach human city planners**

- Blurb

Source: Science https://www.science.org/doi/abs/10.1126/science.aag1520

2013 **Blurb in Science magazine about time travel**

- Blurb

Source: Science https://www.science.org/

2017 **Public lecture: Red Vic lecture series, San Francisco. History and philosophy of Art and Biology.**

- Lecture

Source: verification https://danielarifriedman.com/resume/verify.html

2016 **Public lecture: Red Vic lecture series, San Francisco. History and philosophy of self-organization, stigmergy, and eusocial insects.**

- Lecture

Source: verification https://danielarifriedman.com/resume/verify.html

2021 **Quoted in Dopamine Nation by Dr. Anna Lembke, p. 75: The world is sensory rich and causal poor.**

- Quote

Source: Dopamine Nation https://www.penguinrandomhouse.com/books/624957/dopamine-nation-by-anna-lembke-md/

2025 **No Signal 006: Daniel Ari Friedman and Jakub Smekal || Active Inference**

- Profile or Interview

Source: No Signal https://kernel0x.substack.com/p/no-signal-006-daniel-ari-friedman

2022 **Daniel Friedman: Drawing Decentralized Beauty From Biology**

- Profile or Interview | Curio DAO

Source: Curio DAO Mirror https://mirror.xyz/0x8Fbf5c7Ca1295e892cC865500A8420C3316b889c/ztyis_d2IhlBUAb0LI4mCjIrvomZr939O1T71a5-R1Y

2021 **Written interview with RosyBVM**

- Profile or Interview

Source: Rosy Conversation https://rosybvm.com/2021/01/20/rosy-conversation-with-daniel-ari-friedman/

2019 **Radio Interview, KPOO 89.5 FM in San Francisco**

- Radio

Source: KPOO https://kpoo.com/

Professional Participation, Engagement, and Service (27)

2022-2024 **ISSS | Board of Directors**

- Vice-President for Communication and Systems Education on the International Society for the Systems Sciences Board.

Source: ISSS https://www.isss.org/home/

2021-2023 **JoVE | Academia**

- Co-editor, Field and Laboratory Methods for Modern Entomology.

Source: JoVE methods collection https://www.jove.com/methods-collections/567/field-and-laboratory-methods-for-modern-entomology

2021 **Cybernetics Society | Membership**

- Member (MCybS), Cybernetics Society.

Source: Cybernetics Society http://cybsoc.org/

2020-ongoing	**IPA | Membership** - Information Professional Association. Source: IPA https://information-professionals.org/
2020-ongoing	**Active Inference Institute | Community** - Active Inference Institute co-founder. Source: Active Inference Institute https://www.activeinference.institute/
2020-2023	**Davis Complexity Group | Community** - Davis Complexity Group. Source: verification https://danielarifriedman.com/resume/verify.html
2019-2024	**Complexity Adventures | Community** - Complexity Adventures co-founder. Source: Complexity Adventures https://www.complexityadventures.com/
2018-2021	**ALIUS | Membership** - ALIUS Research Group contributor, editor, and co-organizer. Source: ALIUS Research Group https://www.aliusresearch.org/
2018-2019	**Wonderfest | Community** - Wonderfest Science Envoy; science outreach and community training. Source: Wonderfest https://wonderfest.org/
2017-2021	**ESA | Membership** - Entomological Society of America. Source: Entomological Society of America https://www.entsoc.org/
2017-2019	**Stanford | Academia** - NeuroChoice group, Stanford Neurosciences Institute. Source: NeuroChoice https://neurochoice.stanford.edu/purpose
2017-2018	**Stanford | Membership** - Graduate student member of Arts Committee for Stanford Biology department. Source: Stanford Biology https://biology.stanford.edu/
2016-2019	**Stanford Complexity Group | Community** - Stanford Complexity Group. Source: Stanford https://www.stanford.edu/ | verification https://danielarifriedman.com/resume/verify.html
2016	**Stanford | Community** - Science Teaching Through Art (STAR) outreach program; poster presentations and conversations about ants at local middle and high schools. Source: Stanford https://www.stanford.edu/
2015	**Santa Fe Institute | Workshop** - Santa Fe Institute Complex Systems Summer School, one-month residential program. Source: Santa Fe Institute CSSS https://www.santafe.edu/engage/learn/programs/sfi-complex-systems-summer-school
2014-2018	**Gordon Lab | Field work** - Five summers of ant fieldwork during PhD under Prof. Deborah Gordon at Southwestern Research Station, Arizona. Source: Southwestern Research Station https://www.amnh.org/research/southwestern-research-station
2014	**Stanford | Field work** - Summer field ecology class with Stanford Prof. Rodolfo Dirzo at Las Tuxtlas tropical research station, Mexico. Source: Los Tuxtlas Biological Station https://www.ibiologia.unam.mx/eblt/
2014	**NIMBioS | Workshop** - Summer workshop on Evolutionary and Quantitative Genetics at NIMBioS, University of Tennessee, Knoxville. Source: NIMBioS http://www.nimbios.org/tutorials/TT_eqg

2014	**Science | Field work** - Drosophila work with Dr. Michael Lang in Arizona. Source: Drosophila Evolution https://droseu.net/drosophila-evolution/
2014-ongoing	**Science | Academia** - Invited peer reviewer for 1-10 articles per year across different journals. Source: Science https://www.science.org/
2014-2023	**IUSSI | Membership** - International Union for the Study of Social Insects. Source: IUSSI https://iussi.org/
2014-2017	**Stanford | Community** - Stanford Center for Evolutionary and Human Genomics outreach programs, including teaching genetics in local middle schools and hosting high school students at Stanford. Source: Stanford Center for Evolution and Human Genomics https://cehg.stanford.edu/
2014-2017	**Stanford | Community** - Stanford Bio Core Explorations hands-on lecture/workshop for undergraduates: Art and Biology. Source: Stanford Biology https://biology.stanford.edu/
2013	**UC Davis | Community** - Science mentoring at Woodland High School. Source: Woodland High School https://whs.wjusd.org/
2012	**UC Davis | Community** - Drums and percussion for UC Davis production of RENT. Source: UC Davis https://www.ucdavis.edu/
2012-2013	**UC Davis | Community** - Volunteer in emergency room at hospital in Sacramento, four hours per week for six months. Source: UC Davis Health https://health.ucdavis.edu/
2011-2014	**UC Davis | Community** - Chess Club at UC Davis President (2012-2014), organizing free outdoor chess at the farmer's market twice weekly for two years. Source: UC Davis Campus Recreation https://campusrecreation.ucdavis.edu/

Art Portfolio and Uses (14)

All	**Art portfolio** - Posting drawings to an online portfolio since 2009. Source: Flickr http://flickr.com/photos/daniel_friedman/
2021	**Curio Cards** - A full Curio Cards set sold at Christie's on October 1, 2021 for 393 ETH, approximately $1.2 million. Source: Christie's Curio Cards lot https://www.christies.com/lot/lot-6337619
2021	**Curio Cards** - Curio Cards collection stored in the Arctic World Archive with art and provenance data intended for long-duration preservation. Source: Top Dog Beach Club https://topdogbeachclub.com/non-fungible-vault/
2021	**Curio Cards** - Public auction and market history for Curio Cards includes major 2021 sales; verify venue-specific claims before reusing exact auction-house wording. Source: Sotheby's Curio Cards https://metaverse.sothebys.com/natively-digital/lots/curio-cards-complete-set-1-30-plus-17b
2019	**Art used** - Art used in the eBook Rebelling with Care: Exploring open technologies for commoning healthcare. Source: ResearchGate https://www.researchgate.net/publication/335911369_Rebelling_with_Care_Exploring_open_technologies_for_commoning_healthcare

2019 **Art used**

- Art used in Slate.fr blog.

Source: Slate France https://www.slate.fr/story/173103/fecondation-spermatozoide-ovule-chevalier-belle-au-bois-dormant-mythe-sexisme-naturalisation-biologie?

2019 **Art used**

- Art used in degrowth blog.

Source: Degrowth.info https://www.degrowth.info/blog/on-strategies-for-socioecological-transformation

2018 **Art used**

- Art used in Medium post by Lorena Cabeza, August 23, 2018.

Source: Medium https://medium.com/@lorenacabeza/manifiesto-la-escritura-y-la-vida-10d4c254c0bf

2018 **Art used**

- Cover image of Probe magazine.

Source: Probe Magazine https://www.ssbprobe.com/decentraldogma-danielfriedman

2017 **Curio Cards**

- Early Ethereum art NFT project; art for Curio Cards 24, 25, and 26.

Source: Curio Cards artist https://curio.cards/artist/danielfriedman/ | Curio Cards docs https://docs.curio.cards/the-artists/daniel-friedman

2017 **Art used**

- Art used in Bas Haring blog.

Source: Me Judice https://www.mejudice.nl/artikelen/detail/mijn-economiemuseum-de-markt

2017 **Art used**

- Art used in Vice Motherboard blog.

Source: VICE Motherboard https://www.vice.com/en/article/jpagk8/a-dollar25-million-book-heist-proves-the-printed-page-is-still-sacred

2016 **Art used**

- Featured in Genetics Society of America art blog.

Source: Genes to Genomes https://genestogenomes.org/gsa-art-daniel-friedman/

2016 **Art used**

- Art featured by Stanford Art of Neuroscience competition.

Source: Stanford Scope https://scopeblog.stanford.edu/2016/07/06/art-of-neuroscience-competition-highlights-beauty-of-the-brain/